

moved by surface creep involving rolling (especially for large grains) at relatively low (sub-saltation) wind speeds. However, the total flux may still be far less than that of sediments transported by water on Earth¹⁷. (5) Small-scale surface features: small features transverse to the wind such as ripples and microdunes did not develop in experiments in which transport occurred solely by rolling. However, grooves and ridges parallel to the wind were formed and thus may be expected on Venus. The length of these features was governed by the size of the wind tunnel and it is possible that on Venus they could extend for tens or even hundreds of metres. Depending on the characteristics of radar imaging systems, such as wavelength, and the geometry of the radar beam, radar signatures from the surface of Venus could be influenced by the presence of these grooves and ridges. In these preliminary results, we note that although there is an association between rolling and longitudinal bedforms, the latter features are not necessarily a function of this mode of transport.

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Neutron generation in lightning bolts

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Intense electrical discharges through polymer fibres have been shown¹ to produce 2.45-MeV neutrons, probably by deuteron-deuteron fusion. Noting broad similarities between discharges in polymer fibres and natural lightning, Libby and Leukens² have suggested that neutrons are also generated in lightning flashes, as a result of the fusion of deuterium contained in the atmospheric water vapour; by rescaling the plasma parameters of polymer fibres to those involved in natural lightning, they have predicted a yield of $\sim 10^{15}$ neutrons per lightning flash. An experiment by Fleischer³, however, using fission track detectors placed near lightning arrestors, has failed to ascertain the neutron production in lightning discharges. Based on the number of background cosmic-ray tracks accumulated in these detectors during seven months of observations, Fleischer estimated an upper limit of 2.5×10^{10} neutrons per lightning stroke. In our experiment, we have attempted to keep the cosmic-ray neutron background at a negligible level by searching for neutrons from individual lightning strokes, for a time-interval comparable with the duration of the lightning stroke. Here we present the first experimental evidence that neutrons are generated in lightning discharges, with 10^7 - 10^{10} neutrons per stroke. Whether these neutrons are thermonuclear in origin, or are generated by non-thermal processes, remains to be determined.

Neutrons were detected at Gulmarg (altitude 2,743 m, latitude 34.07° N, longitude 74.42° E) by a low-energy cosmic-ray

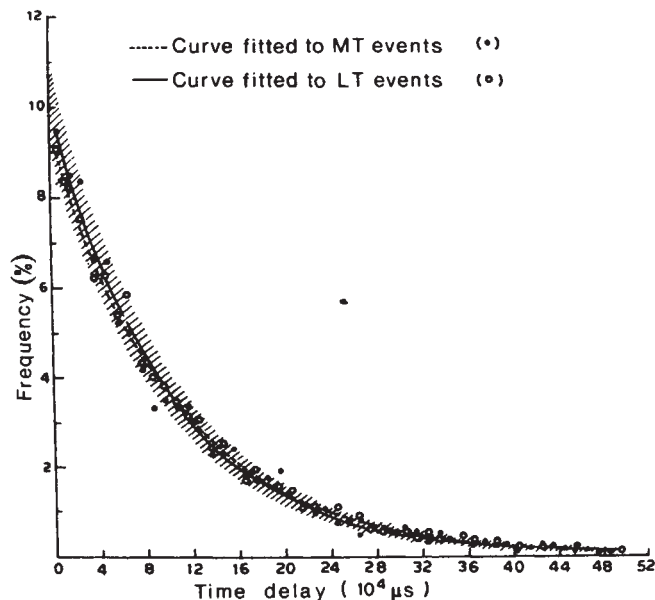


Fig. 1 The time-delay distributions of one-neutron lightning-triggered (LT) and manually-triggered (MT) events. Exponential curves fit both the data sets and produce a mean count rate of 10 neutrons s^{-1} , suggesting that these events are related to the background cosmic rays. The hatched portion represents the error range estimated for the fitted curves.

neutron monitor comprising 21 enriched BF_3 counters, each 90 cm long and 3.8 cm in diameter. The counters are arranged in the form of a pile and are placed over 28-cm-thick paraffin wax slabs 8 m above the ground, which helps render the monitor essentially opaque to low-energy background neutrons entering it from below. The counters are also covered from above with 7.5-cm-thick paraffin wax to thermalize 2.45-MeV neutrons and allow their detection with an efficiency of $\sim 3\%$. The monitor has an effective surface area of $3 \times 10^4 \text{ cm}^2$ and continuously records low-energy cosmic-ray background neutrons with an average rate of 1.8×10^4 neutrons per 30 min. The pressure coefficient for the monitor is -0.66% per mbar, in close agreement with reported values^{4,5}.

A linear antenna is mounted in the vicinity of the monitor to sense the short-term electric field variation associated with a lightning stroke. The signal, after suitable amplification and amplitude discrimination, activates an interval-counter to count 10- μs pulses from an oscillator, from the instant of the sensing of the electric field to the arrival of the first neutron detected in the pile. The time delay, thus registered in the arrival of the prompt neutron, is used for computing the distance to the lightning stroke, assuming a line-of-sight propagation. This neutron and those detected within the next 320 μs are recorded in four counters, each opened sequentially for an interval of 80 μs and having provision for counting a maximum of 99 neutrons. At the completion of the recording sequence, the time delay and the neutron counts are printed and the system primed for a subsequent lightning event after a dead-time of 400 ms. Note that the total counting time of 320 μs is large compared with the 50- μs duration of a typical lightning stroke and small compared with the average inter-stroke time of 40 ms⁶. The cosmic-ray background is negligible in this time and, thus, it is possible to monitor neutrons from individual lightning strokes with a high signal-to-noise ratio.

Between May 1980 and May 1983, the Gulmarg experiment has responded 11,200 times to potential gradient variations from lightning; it has also been activated manually on 8,400 occasions by feeding a pulse generator output to the lightning channel discriminator unit at randomly chosen epochs. Table 1 compares the frequency of the lightning-triggered (LT) events as a function of neutron number recorded per event with the corresponding

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number of manually triggered (MT) events. It is evident that, in a large majority of both LT and MT events, only one neutron is recorded within the monitoring interval of 320 μ s. Figure 1 shows, for these event groups, the distribution of the time-delays as recorded by the interval counter. Exponential curves fitting both data sets do not differ significantly from one another, suggesting a random arrival time of the detected neutrons in both cases. Further, the distributions result in an average time delay of 0.1 s between the arrival of successive neutrons, in agreement with the count rate of 10 neutrons s^{-1} recorded by the monitor for the background cosmic rays. These observations demonstrate clearly that the single neutron events result from the random arrival of cosmic-ray neutrons in the monitor and have no association with the preceding lightning discharge. The two-neutron events (see Table 1) are observed to constitute 2.23% and 1.11% of the total observed LT and MT events, respectively. These numbers are higher than the values calculated for the chance arrival of two neutrons from the cosmic-ray background, within a recording time of 320 μ s. It is conceivable that some of the two-neutron events result from the neutron bunches expected to be generated by the interaction of cosmic-ray hadrons of energy $\geq 10^{11}$ eV with the material present around the pile. At the same time, a possible association of these events with lightning flashes is suggested by the greater percentage of two-neutron LT events observed compared with the corresponding MT events.

Only for events of three or more neutrons are major differences observed between lightning and manual triggers. There are only four MT events with three neutrons and none with more than three neutrons. By contrast, 124 LT events have been registered with ≥ 3 neutrons, including 59 and 39 events with ≥ 6 and ≥ 9 neutrons, respectively, and one with a maximum count of 60 neutrons. The large differences in the relative number of LT and MT events and in the number of constituent neutrons show that most of the LT events with ≥ 3 neutrons cannot arise in the same manner as the four corresponding MT events. The processes precluded include neutrons from the cosmic-ray background, produced either singly, by low-energy primaries, or in bunches, by high-energy hadron interactions. The association of recorded neutrons with the cosmic-ray shower secondaries of primary energy $\geq 3 \times 10^{13}$ eV is also precluded, as only single neutron events were recorded when, in the course of a simultaneous run for 48 h, the system was triggered $\sim 2,800$ times by the output of a 10 m $\times$ 10 m shower array. The shower array boundary was ~ 4 m from the centre of the neutron pile and a finite number of shower triggerings should have been followed by a significant neutron count in the pile. That only single neutron events were recorded indicates the relative insensitivity of the neutron pile to high-energy shower neutrons. As the whole experiment is fully powered by batteries, any lightning-induced transients in the power lines are not expected to affect the operation of the pile. That the half-hourly counting rates of the monitor do not show any erratic behaviour further suggests that the lightning-generated electrical noise does not also adversely affect the operation of the system electronics, including the data recording unit. Moreover, it is not possible in this way to account for ≥ 3 -neutron events with time delays of 10 – 10^3 μ s, as the lightning-associated electrical noise should generally affect the recording system almost simultaneously with the activation of the lightning detector and thus be recorded as events with smaller time delays. That this is not the case, is clearly shown by our

data wherein, out of 124 LT events with ≥ 3 neutrons, 56 LT events have time-delays in the range 10^2 – 10^5 μ s.

The possibility of electromagnetic pick-up originating from subsequent strokes rather than from the first return stroke has been considered, although this is unlikely to happen, as the first stroke is usually the strongest stroke in a flash. One cannot, *a priori*, preclude the possibility of the system being triggered by strong electromagnetic activity preceding a lightning flash, whereas the neutron signal may be caused by pick-up from the lightning stroke. We have, therefore, verified experimentally how susceptible our system is to electromagnetic pick-up under natural lightning conditions. For this purpose, additional logic circuitry has been incorporated in the system, which is capable of detecting a maximum of four succeeding strokes of a lightning flash and recording their relative time spacings. This differential time information, together with the provision for recording the delay between the initial triggering of the system and the arrival of neutrons in four time-gates of 80 μ s each, would then reveal any direct connection between the neutron events and the lightning strokes. The experiment has been operated with this modification for about three months and several hundred LT events have been recorded, many of which were multi-stroke events. Whereas most of them were one-neutron events, a total of 24 LT events were two-neutron events and two LT events had three neutrons each. A comparison of the integrated and differential time information for the events reveals that in none were the detected neutrons in close coincidence with the detection of electromagnetic pulses from a single or a multi-stroke lightning flash, which reinforces our contention that the neutron signal cannot be attributed to any lightning-induced transients. It is therefore logical to associate the 124 LT events having ≥ 3 neutrons with neutron generation in lightning and to interpret the observed time-delays as the propagation time of these neutrons from the location of the lightning discharges.

Assuming a simple geometry consisting of a vertical lightning channel of length H equal to 2 km, at a distance r from the detector, the following approximate relation has been used to calculate the neutron yield per stroke

$$N = (N_D / \eta A) [(4\pi H r) / (\tan^{-1} H r)] (\exp \Sigma r) \quad (1)$$

where r (cm) = $13.8 \times 10^5 \sqrt{E_n}$, N is the total number of neutrons produced in the lightning channel, N_D the number of transmitted neutrons recorded by the neutron monitor, and η and A , the detection efficiency and the effective area of the monitor respectively. The total macroscopic cross-section, Σ , at the observation site, for neutrons of energy E_n (eV) is computed from the published microscopic scattering and absorption cross-section values⁷ for an atmospheric composition of 80% nitrogen, 20% oxygen.

Figure 2 shows the time-delay, t , and the number of neutrons recorded during the sampling time of 320 μ s for each of the 124 LT events. Evidently, 50 events have time delays in the range 10 – 50 μ s, which, if the constituent neutrons are assumed to be of 2.45-MeV energy, result in a total yield of 9×10^6 – 2×10^{10} neutrons per stroke. Such events would pass undetected in the measurements of Fleischer³, because of the average detection limit of 2.5×10^{10} neutrons per stroke inherent in his mode of detection. The time delays recorded for the remaining 65 events are well spread between 60 and 2×10^5 μ s and result in neutron yields too large to have gone undetected in earlier experiments. In addition, such high fluxes are also incompatible with the physical conditions obtained in a lightning channel and necessitate a reinterpretation of the large time delays exhibited by these events. It is possible that these events originated in multi-stroke lightning flashes; the first stroke triggers our system, but the neutrons are generated in a succeeding stroke belonging to the same or a different flash. Another plausible explanation is that these events comprise neutrons of lower energy produced⁸ other than by the postulated deuteron-deuteron fusion reaction, for example by $^{12}\text{C}(\text{H}, \text{n})^{13}\text{N}$ and $^{14}\text{N}(\text{H}, \text{n})^{15}\text{O}$. It is also worth considering that, when lightning strikes a tree, neutrons may be generated by the fusion of deuterons present in the vaporized

Table 1 Manually triggered (MT) and lightning-triggered (LT) events with different neutron numbers

No. of neutrons	No. of events	
	MT type	LT type
1	8,299 (98.84%)	10,818 (96.65%)
2	97 (1.11%)	250 (2.23%)
3	4 (0.05%)	40 (0.35%)
>3	0 (0.00%)	84 (0.77%)

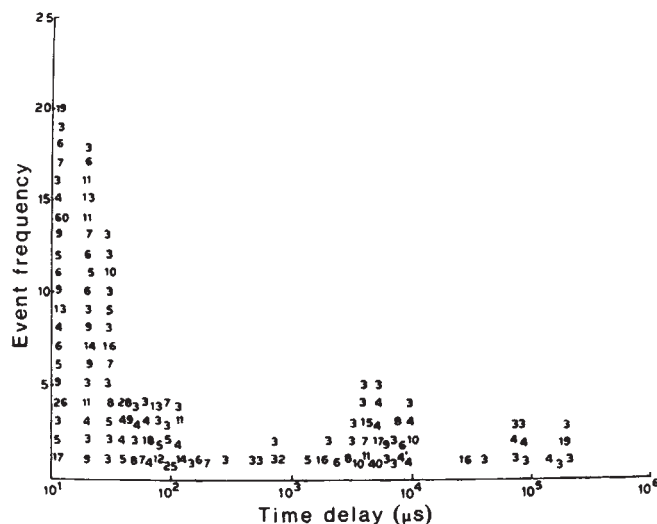


Fig. 2 The time-delay distribution of the 124 lightning-triggered (LT) events having ≥ 3 neutrons. The digits represent the number of constituent neutrons recorded in the sampling time of 320 μ s.

wood of the tree⁹. Such neutrons would undergo substantial energy degradation by the surrounding wood before reaching the detector and would be registered as events with large time delays.

The reason for considering these alternatives seriously becomes apparent when considering the following two events recorded at Gulmarg: a lightning is known to have struck a tree, 1.5 km away, resulting in the registering of 10 neutrons after a time delay of 30 μ s. Even after correcting for the delay in sensing the electric field variation due to its finite rise-time, the actual distance for this event is still underestimated by almost a factor of two when neutrons of energy equal to 2.45 MeV alone are considered. One may infer, therefore, that the neutron energy was > 2.45 MeV. On the other hand, the more plausible explanation for the small time delay is that the first detected neutron is a randomly-arriving cosmic-ray background neutron followed subsequently by neutrons from the lightning channel after a time lag.

In another instance, a fir tree, situated ~ 400 m away from our detector, was severely damaged in a lightning strike in which 33 neutrons were recorded in 320 μ s. The observed time delay of $7.174 \times 10^4 \mu$ s for this event places the source at a distance several orders of magnitude farther than the actual value of 400 m, assuming a neutron flux of 2.45-MeV energy. On the other hand, it is possible to account for the observed distance properly if we assume neutrons to have an energy as low as 0.2 eV. In fact, choosing neutrons with energies between 10^3 and 0.023 eV and accounting for the atmospheric absorption, we can reconcile the large time delays observed for 65 events with plausible yields of $3 \times 10^6 - 2 \times 10^{12}$ neutrons per stroke. We note that these calculations assume a continuous emission of neutrons along the lightning channel, which may not necessarily be true. Instead, neutrons may be generated in a limited length of the channel where conditions are most favourable. In the extreme situation of neutron generation from a point source, however, the calculated flux values do not change substantially from those of a line source. Furthermore, we must emphasize that the above estimates are only approximate and that, to calculate the neutron yield and interpret the observed time delay more accurately, one should consider atmospheric scattering and absorption effects under stormy conditions. Detailed modelling, using more accurate neutron propagation theories, is required for this purpose.

In conclusion, we have presented here evidence of possible neutron generation in lightning discharges, at yields of $\sim 10^7 - 10^{10}$ neutrons per stroke. That this is several orders of magnitude lower than the predicted values² is partly explained on the basis

that, while rescaling the exploding wire results to the lightning process, the significant differences in the respective plasma temperatures and reaction cross-section for the neutron generation were ignored. Further, it is not clear from our experiment whether these neutrons are thermonuclear in origin or are generated by non-thermal processes. In future, it will be important to obtain the lightning source distance independently to overcome the uncertainties, inherent in our technique, in estimating the yields from the recorded 'time delays'. There should also be provision for recording the lightning-generated neutron events simultaneously at another pile installed at an appropriate distance. A boron or cadmium cover placed over the pile would then further establish the genuineness of the events and broadly determine the energy of the neutrons.

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When does small-scale convection begin beneath oceanic lithosphere?

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Convection on a scale smaller than the horizontal dimensions of lithospheric plates can be produced by instabilities in the upper or lower boundary layer of the larger mantle flow. Small-scale convection associated with the upper boundary layer should produce a detectable gravity signal and affect the rate of cooling and subsidence of the lithosphere. I describe here a numerical model of small-scale convection in a fluid of variable viscosity. The results indicate that recently observed gravity anomalies¹ showing a pattern of highs and lows aligned in the direction of oceanic plate motion may be the result of small-scale mantle flow. The convective flow must begin in the first 6 Myr of lithospheric cooling to produce the observed signals, which is not inconsistent with constraints on the viscosity of the mantle. The calculated trend for the subsidence of the ocean floor is found to be almost linear with (time)^{1/2} even when small-scale convection has significantly changed the rate of subsidence. For average shallow asthenospheric viscosities of $\sim 10^{18}$ Pa s, the model subsidence can match data for the oceans^{2,3} and reproduce the magnitude and wavelength of the observed gravity anomalies.

The cooling of the oceanic lithosphere as it moves away from a ridge will be affected by small-scale convection. These effects are studied by calculating the flow in two-dimensional boxes induced by the cooling of a fluid from above. This two-dimensional treatment is justified by experiments on the interaction between two scales of convection, of which the smaller-scale flow is in the form of two-dimensional rolls⁴ where axes are oriented parallel to the larger flow, analogous to the plate

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